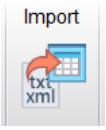
GraphPad Prism introductory activities

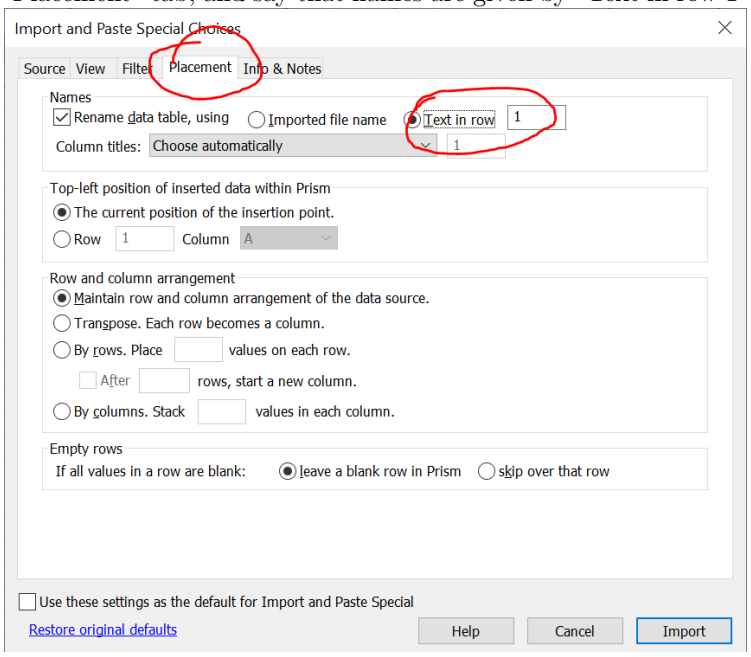
Here are a series of activities you can work through to become more familiar with the plots that can be produced and analyses that can be undertaken using GraphPad Prism.

Comparing the mean of two groups

The dataset to use here is `fig4F_data.csv`.

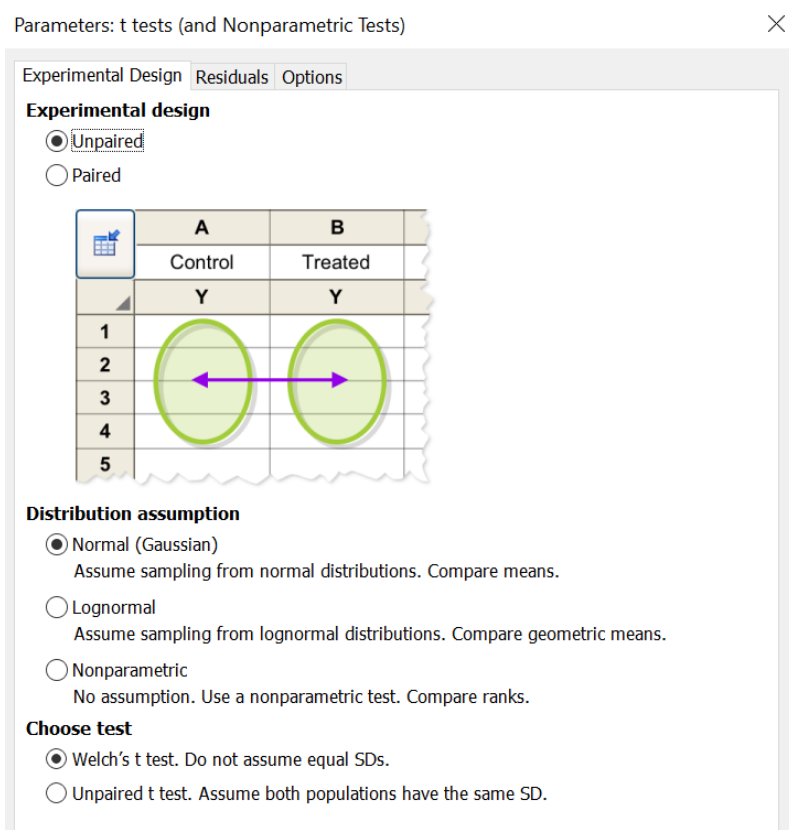
- Create a new Prism dataset as a "Column table".
  - Use the "Enter replicate values, stacked into columns" option
- Import the data:
  - The easiest way is to open the dataset in Excel, and then copy it directly in
  - Or you can click on the "Import" button 

, or click on the "File" menu and then "Import…"
    * Select the `fig4F_data.csv` file
    * In the "Import and Paste Special Choices" window, go to the "Placement" tab, and say that names are given by "Text in row 1"



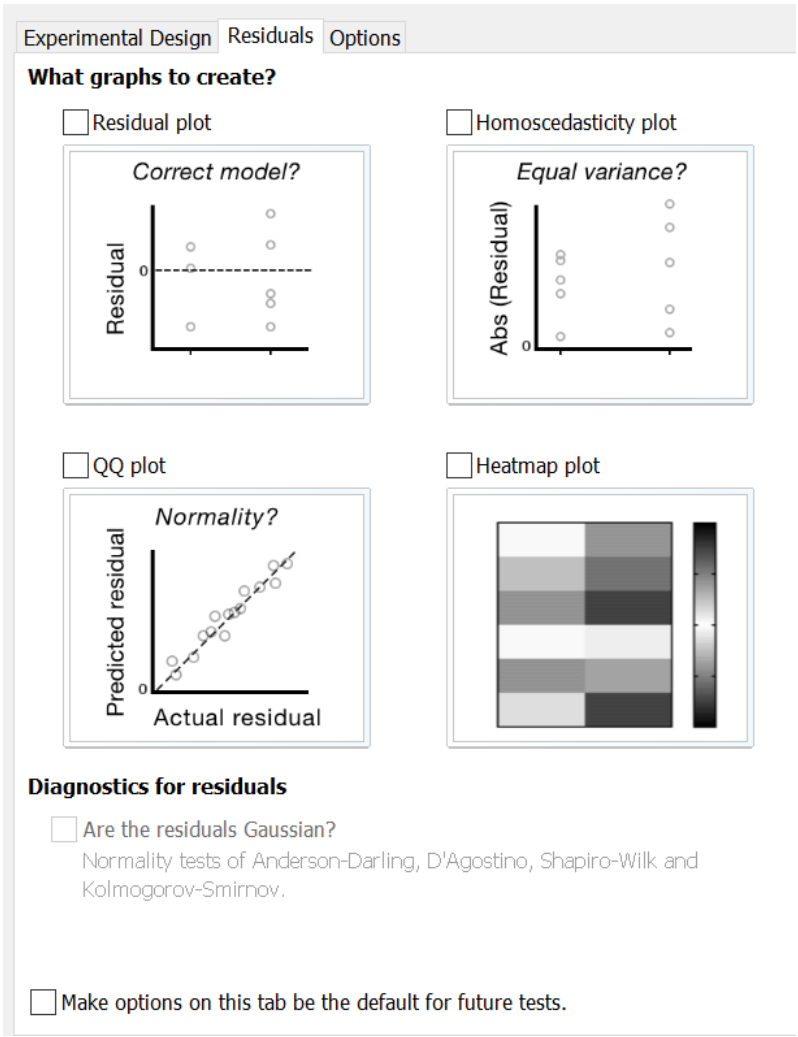
  - Note: You will see some values with a "*" after the number. Cell entries that are not purely numbers will be treated as *missing data*.
- Produce descriptive statistics. You can do this by:

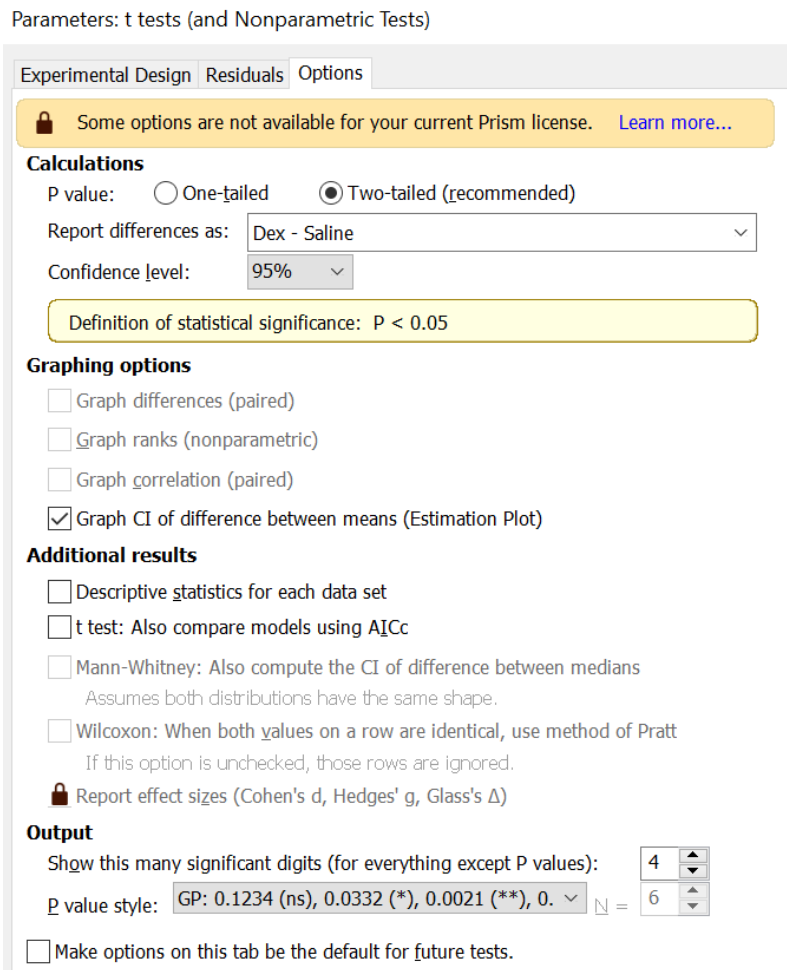
  - Clicking on the “Analyze” button (top ribbon), and then select “Descriptive Statistics” under “Column analyses”
  - Clicking on the “New analysis” (left panel), and then select “Descriptive Statistics” under “Column analyses”
  - Go through the “Analyze” menu, then “Data Exploration and Summary” and then “Descriptive Statistics”
- (optionally) If you want to check for normality within groups then:
  - Clicking on the “Analyze” button (top ribbon), and then select “Normality and Lognormality tests” under “Column analyses”
  - Choose just *one* of the normality tests (e.g., the D’Agostino-Pearson test); deselect the others.
- Compare the mean of two groups:
  - Clicking on the “Analyze” button (top ribbon), and then select “t-tests (and non-parametric tests)” under “Column analyses”
  - In this case, the default options are OK: the data are *not* paired; assume a *normal distribution*, do *not* assume that the two standard deviations are equal.



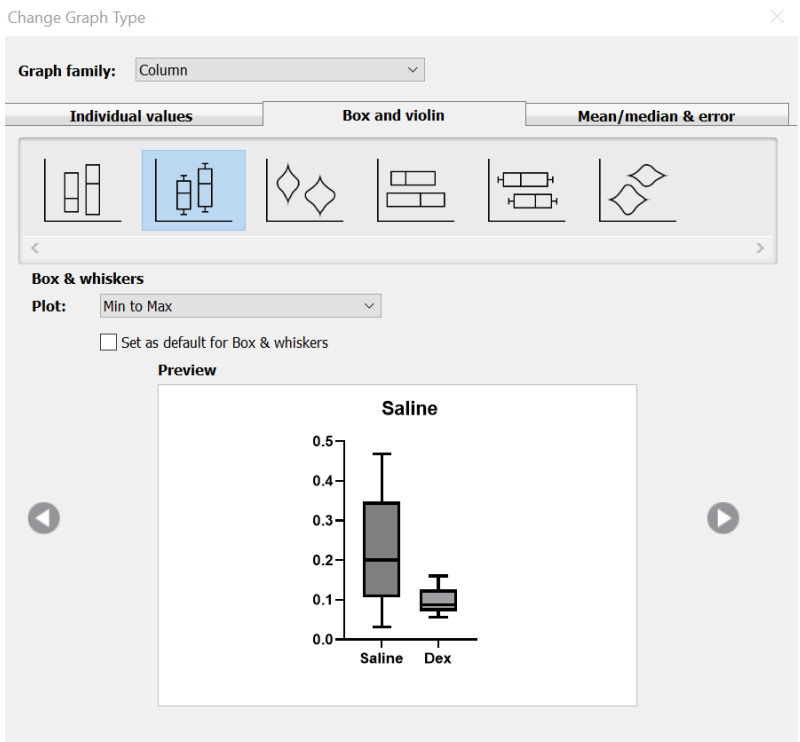
  - Check out the other two tabs for options for reporting of results and checking of assumptions.

Parameters: t tests (and Nonparametric Tests)

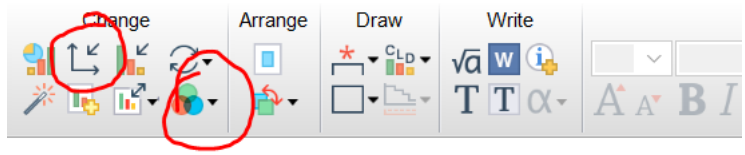


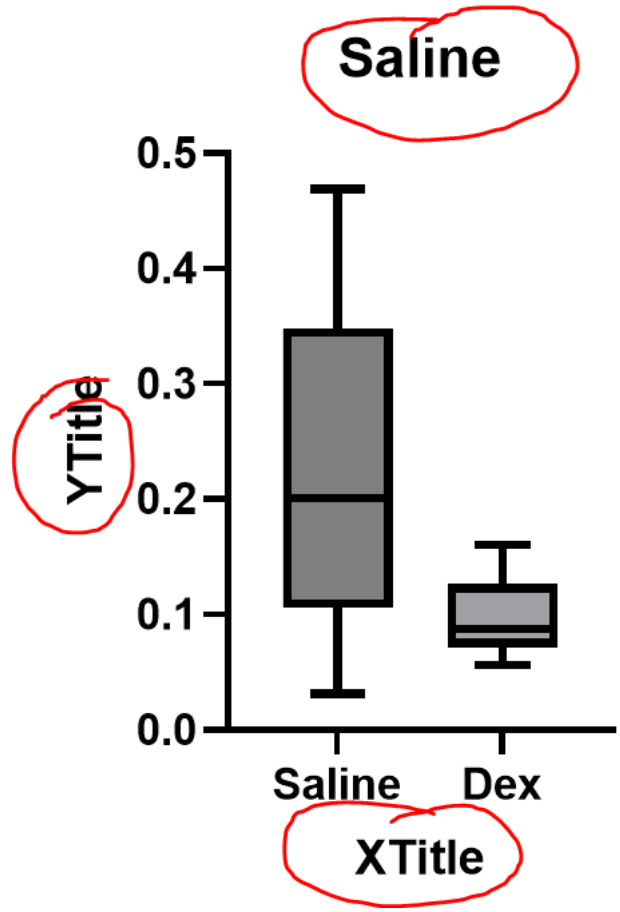


  - (optionally) You can run the analysis again, with different options to see how sensitive the conclusions are to the assumptions of the test.
- Plotting the distribution of the two groups:
  - Either click on the existing entry under "Graphs" (left panel) that corresponds to the name of the "Column table" dataset you created, *or* click on "New Graph" in the left panel.
  - Select a Box and Whisker type plot in the pop-up window



– Select the "Adjust Colors" button to change the colour scheme, the "Adjust Axis" button to modify the axes, and the axis titles and figure





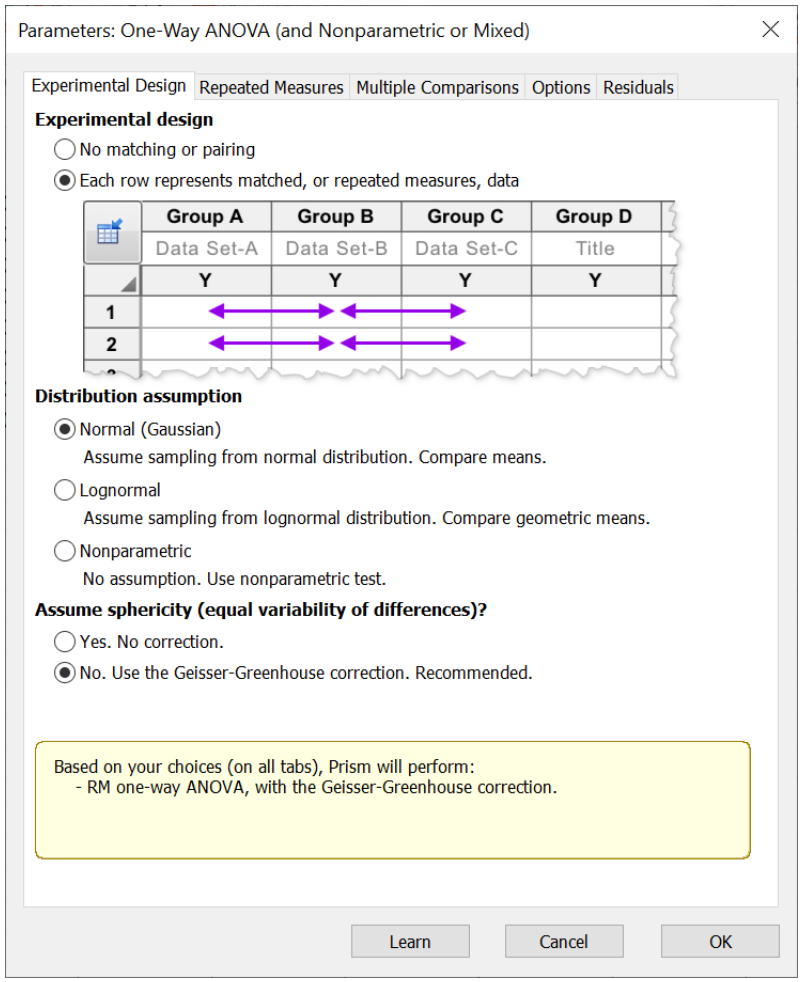
title to change the text.

Comparing the mean of more than two groups

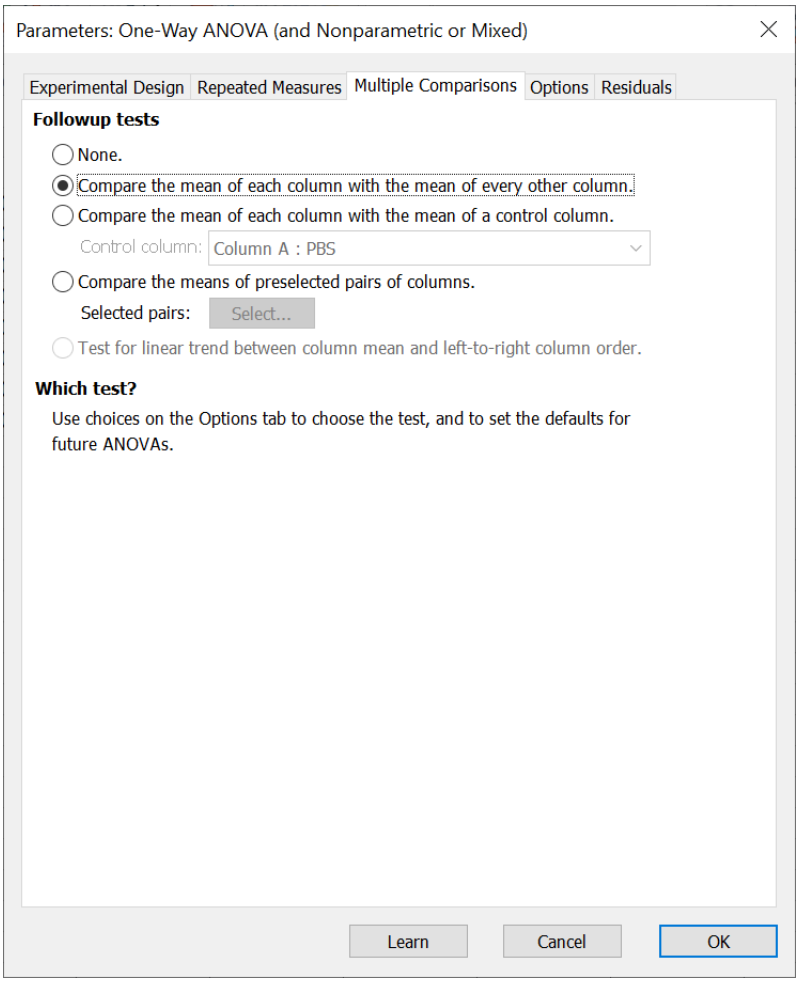
The dataset to use here is `Fig1B_data.xlsx`; data for the three panel plots are found in three different tabs/worksheets within the file.

- Create a new Prism dataset as a "Column table".

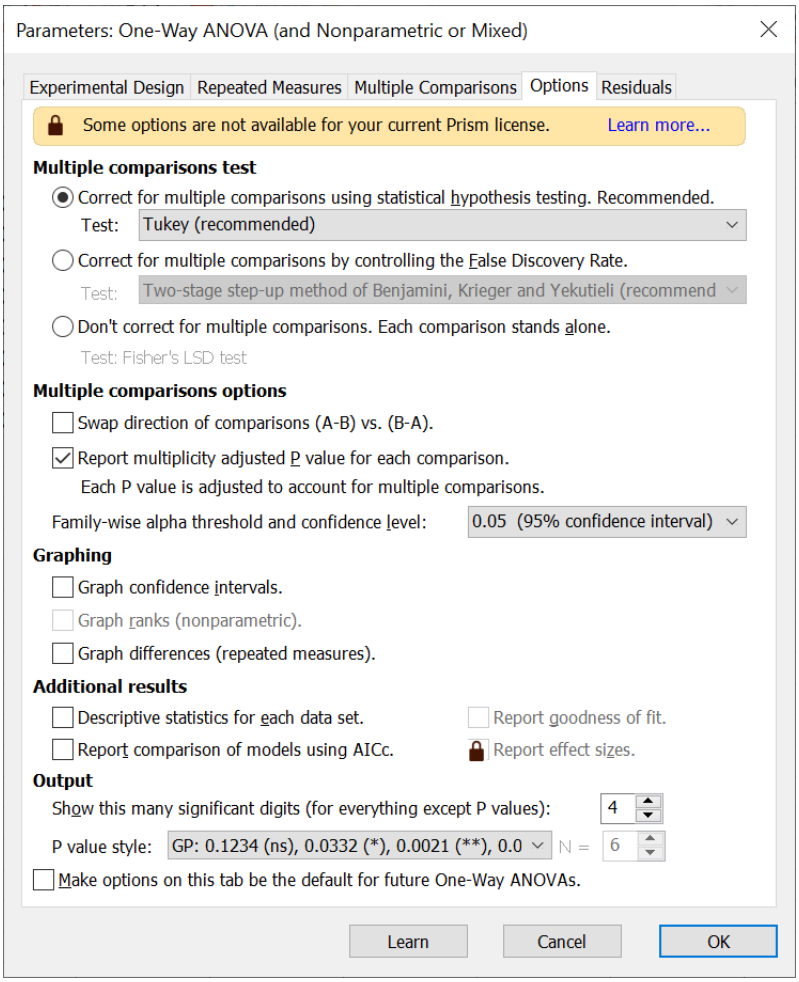
  - Use the “Enter paired or repeated measures data - each subject on a separate row” option
- Import the data:
  - The easiest way is to open the dataset in Excel, and then copy it directly from one of the tabs.
  - Copy the data from each tab into its own “Column table” (so you should end up with *three* datasets within the one Prism project)
  - Rename each of the three datasets. Right click on the dataset name in the left panel, select “Rename dataset”
- Produce descriptive statistics. You can do this by:
  - Steps are as above
- (optionally) If you want to check for normality within groups then:
- Compare the mean of more than two groups
  - Clicking on the “Analyze” button (top ribbon), and then select “One-way ANOVA (and nonparametric or mixed)” under “Column analyses”
  - A window box will pop up - make sure that the options are appropriate (unpaired or paired, parametric/normal or nonparametric, assuming equal variance or not)

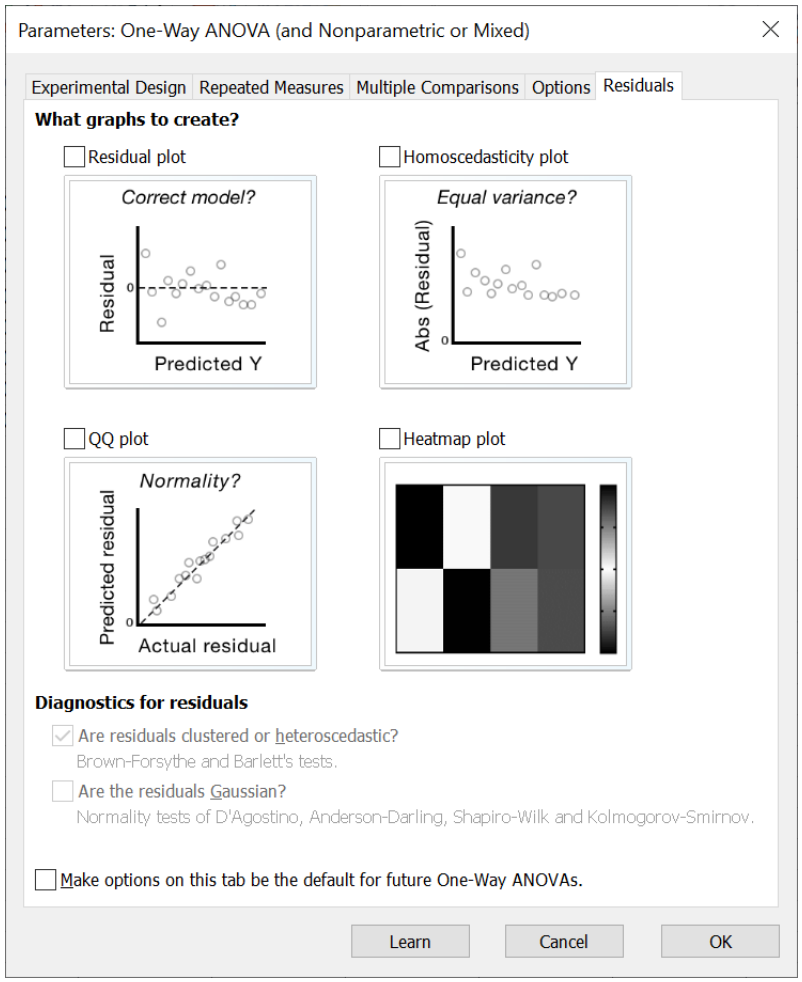


– You will probably want to compare the means between groups. Review the options under the "Multiple comparisons" tab

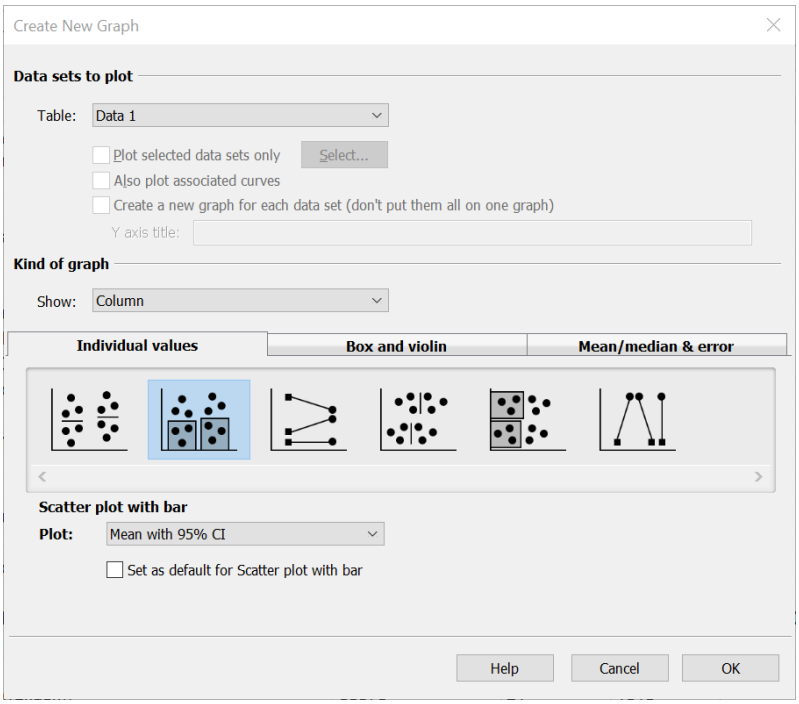


– Review options for reporting of results and checking assumptions, under the “Options” and “Residuals” tabs


Parameters: One-Way ANOVA (and Nonparametric or Mixed)
Experimental Design
Repeated Measures
Multiple Comparisons
Options
Residuals
Some options are not available for your current Prism license.
Learn more...
Multiple comparisons test
Correct for multiple comparisons using statistical hypothesis testing. Recommended.
Test:
Tukey (recommended)
Correct for multiple comparisons by controlling the False Discovery Rate.
Test:
Two-stage step-up method of Benjamini, Krieger and Yekutieli (recommend
Don't correct for multiple comparisons. Each comparison stands alone.
Test: Fisher's LSD test
Multiple comparisons options
Swap direction of comparisons (A-B) vs. (B-A).
Report multiplicity adjusted P value for each comparison.
Each P value is adjusted to account for multiple comparisons.
Family-wise alpha threshold and confidence level:
0.05 (95% confidence interval)
Graphing
Graph confidence intervals.
Graph ranks (nonparametric).
Graph differences (repeated measures).
Additional results
Descriptive statistics for each data set.
Report goodness of fit.
Report comparison of models using AICc.
Report effect sizes.
Output
Show this many significant digits (for everything except P values):
4
P value style:
GP: 0.1234 (ns), 0.0332 (*), 0.0021 (**), 0.0
N =
6
Make options on this tab be the default for future One-Way ANOVAs.
Learn
Cancel
OK



  - (optionally) You can run the analysis again, with different options to see how sensitive the conclusions are to the assumptions of the test.
- Plotting the distribution of the two groups:
  - Either click on the existing entry under "Graphs" (left panel) that corresponds to the name of the "Column table" dataset you created, *or* click on "New Graph" in the left panel.
  - In the pop-up window, select a Column type plot with individual values shown, with the bar and whiskers corresponding to the mean and its 95% CI



  - Adjust plot elements as required
  - (optionally) Once you have repeated this process, creating plots for three datasets, you can set up a new "Layout" and set up a panel plot

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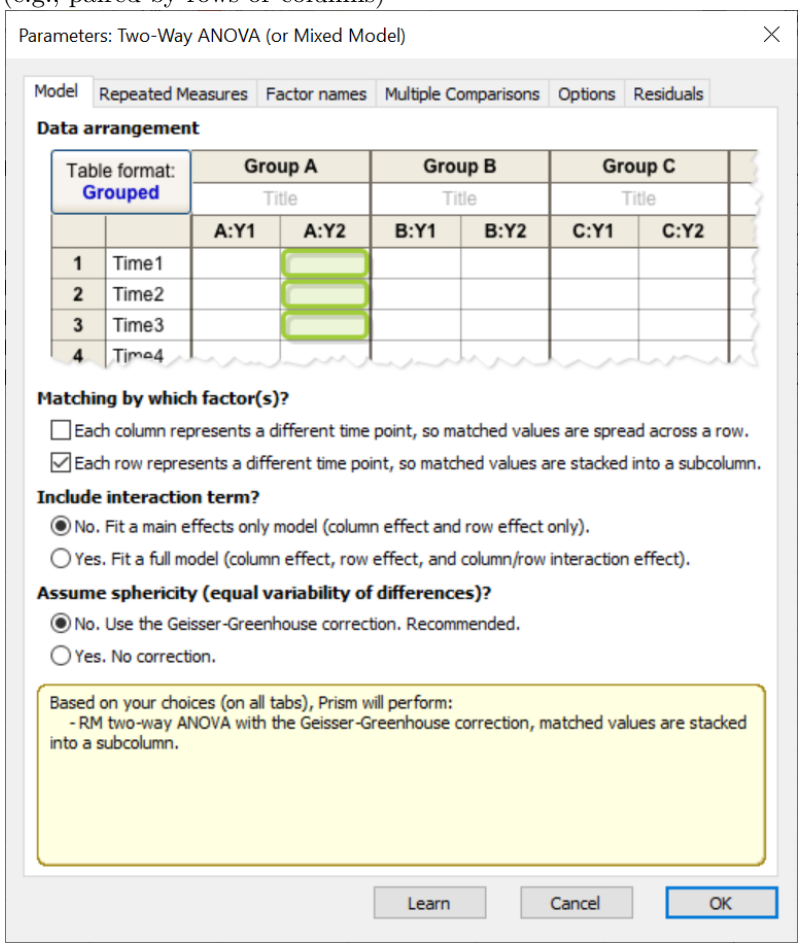
Comparing group means for two factors (two-way ANOVA)

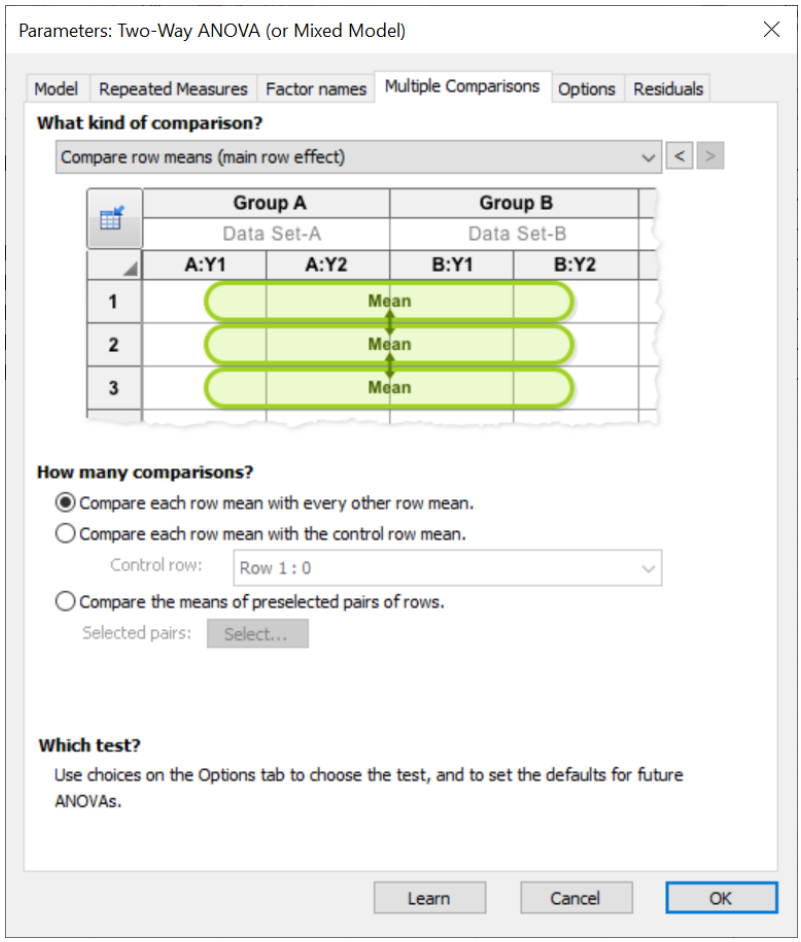
The dataset to use here is `Fig4A_data.xlsx`; the same data is organised in three different tabs/worksheets within the file. The first of these is Prism's preferred way of inputting the data into a Grouped table. You can also consider the challenge of rearranging the data from the formats provided in the second and third tabs. It is easiest to do this in Excel.

- Create a new Prism dataset as a "Grouped table".
  - Use the "Enter *6* replicate values in side-by-side coluns" option
- Import the data:
  - The easiest way is to open the dataset in Excel, and then copy it directly from one of the tabs.
  - Make sure that you copy the group names separately, as well as the row titles (in this case, the number of weeks post injection)
  - Rename the three dataset to give it a meaningful name
- Produce descriptive statistics. You can do this by:
  - Click on "New analysis" (left panel), then under "Grouped analyses" select "Row statistics"

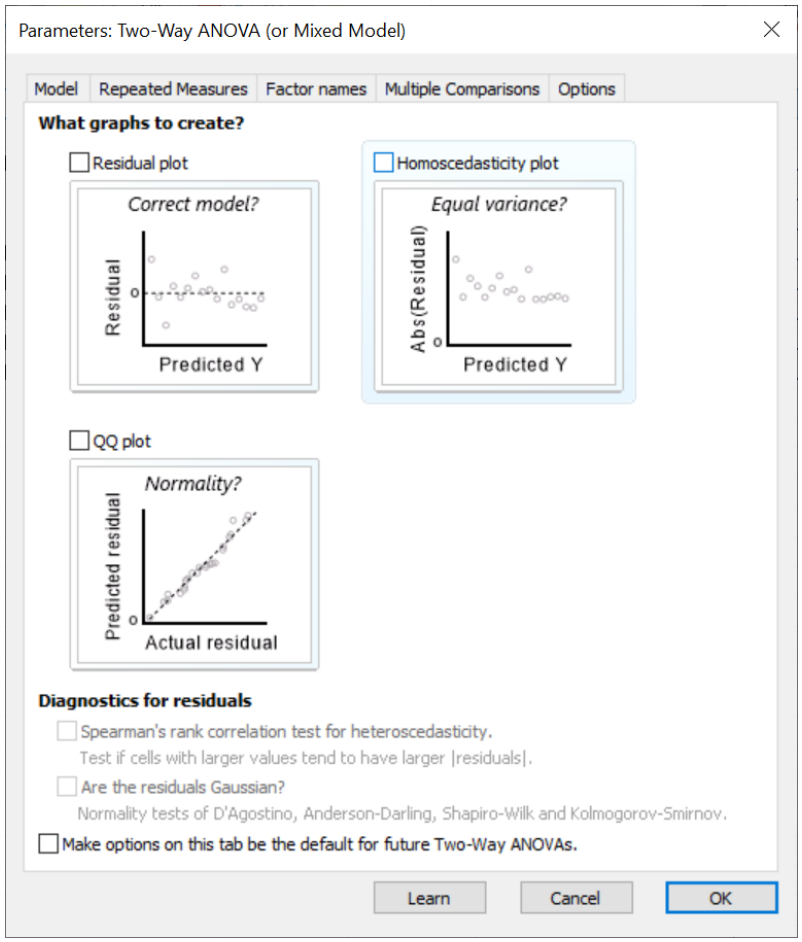
  - You then get to select if you want row or column means; choose row means.
- Compare the mean of two factors simultaneously
  - Clicking on the "Analyze" button (top ribbon), and then select "Two-way ANOVA (or mixed model)" under "Grouped analyses"
  - A window box will pop up - make sure that the options are appropriate (e.g., paired by rows or columns)



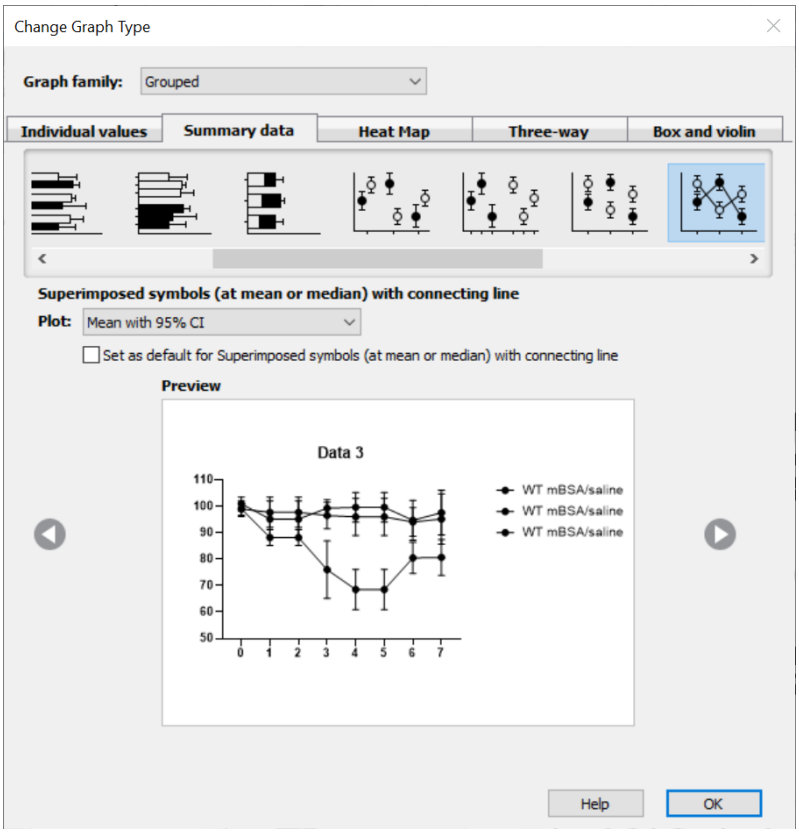
  - You will probably want to compare the means between groups. Review the options under the "Multiple comparisons" tab



– Review options for reporting of results and checking assumptions, under the “Options” and “Residuals” tabs



  - (optionally) You can run the analysis again, with different options to see how sensitive the conclusions are to the assumptions of the test.
- Plotting the distribution of the two groups:
  - Either click on the existing entry under "Graphs" (left panel) that corresponds to the name of the "Column table" dataset you created, *or* click on "New Graph" in the left panel.
  - In the pop-up window, select a Column type plot with individual values shown, with the bar and whiskers corresponding to the mean and its 95% CI

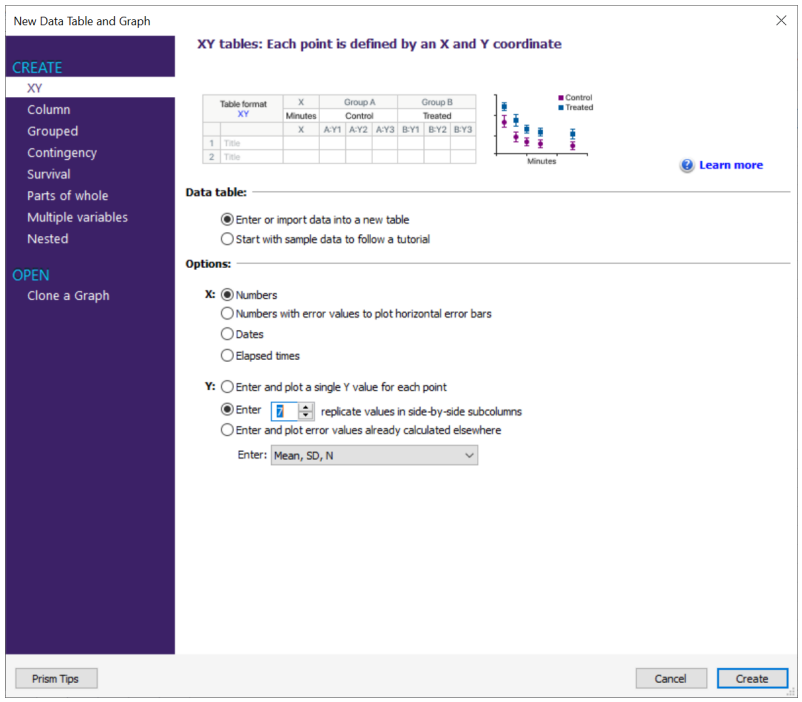


– Adjust plot elements as required

Producing a scatterplot with trendline

The dataset to use here is `penguins.csv`, but the specific extract to plot, which may be easier to work with is in `penguins_adelie_wide_massVsSexVsFlipperLength.csv`.

- Create a new Prism dataset as a "XY table"
  - The original data are provided in `penguins.csv`, but it requires a fair bit of reorganisation. It will be easier to open the `penguins_adelie_wide_massVsSexVsFlipperLength.csv` file in Excel.
  - Indicate that the X values are numbers and for the Y values, "Enter 7 replicate values in side-by-side columns"



Correlation analyses

The dataset to use here is `MetabolicRate.csv`

- Create a new Prism dataset as a "Multiple variables table"